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Electrical connector assembly with connector rotating unit

Abstract

An electrical connector assembly includes a housing and a connector rotating unit. The connector rotating unit is rotatable relative to the housing under the action of an external force. The connector rotating unit includes a rotating body, a movable post and an electrical connector. The electrical connector assembly also includes an arc-shaped limiting groove mated with the movable post and an elastic member mated with the movable post. One end of the elastic member is fixed relative to the housing, and another end of the elastic member is connected with the movable post. The movable post is movable in the arc-shaped limiting groove. The elastic member is configured to rotate the connector rotating unit back to an original position after the connector rotating unit is rotated.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

(1) This patent application claims priority of a Chinese Patent Application No. 202210882908.4, filed on Jul. 26, 2022 and titled “ELECTRICAL CONNECTOR ASSEMBLY”, the entire content of which is incorporated herein by reference.

TECHNICAL FIELD

(2) The present disclosure relates to an electrical connector assembly, which belongs to a technical field of connectors.

BACKGROUND

(3) An electrical connector assembly in the related art includes a cable, an electrical connector connected to the cable, and a housing located outside the electrical connector. In some applications, the electrical connector needs to be able to rotate at a certain angle to better mate with a mating connector. However, how to return the electrical connector back to its original position is a technical problem to be solved by those skilled in the art.

SUMMARY

(4) An object of the present disclosure is to provide an electrical connector assembly with a function of returning to an original/normal position.

(5) In order to achieve the above object, the present disclosure adopts the following technical solution: an electrical connector assembly, including: a housing; and a connector rotating unit, the connector rotating unit being rotatable relative to the housing under an action of an external force, the connector rotating unit including a rotating body, a movable post and an electrical connector fixedly connected with the rotating body, the electrical connector at least partially extending beyond the housing; wherein the electrical connector assembly is further provided with an arc-shaped limiting groove mated with the movable post and an elastic member mated with the movable post; one end of the elastic member is fixed relative to the housing, and another end of the elastic member is connected to the movable post; the movable post is movable in the arc-shaped limiting groove; the elastic member is configured to drive the movable post by a restoring force of the elastic member after the connector rotating unit is rotated, so as to rotate the connector rotating unit back to an original position.

(6) In order to achieve the above object, the present disclosure adopts the following technical solution: an electrical connector assembly, including: a housing including a first housing and a second housing, the first housing defining a first accommodating space, the second housing defining a second accommodating space; and a connector rotating unit, the connector rotating unit being rotatable relative to the housing under an action of an external force, the connector rotating unit including a rotating body, a movable post and an electrical connector fixedly connected with the rotating body, the rotating body being at least partially received in the first accommodating space and the second accommodating space, the electrical connector at least partially extending beyond the housing; wherein the electrical connector assembly is further provided with a limiting groove and an elastic member; one end of the elastic member is fixed relative to the housing, and another end of the elastic member is connected to the movable post; the movable post is movable in the limiting groove when the connector rotating unit is rotating; the elastic member is configured to drive the movable post by a restoring force of the elastic member so as to rotate the connector rotating unit back to an original position.

(7) Compared with the prior art, the present disclosure is provided with a movable post capable of moving in the arc-shaped limiting groove. The elastic member is configured to automatically rotate

the connector rotating unit back to the original/normal position by causing the restoring force of the elastic member to drive the movable post after the connector rotating unit is rotated.

Description

BRIEF DESCRIPTION OF DRAWINGS

- (1) FIG. 1 is a schematic perspective view of an electrical connector assembly in accordance with an embodiment of the present disclosure;
- (2) FIG. 2 is a front view of FIG. 1;
- (3) FIG. 3 is a partial perspective exploded view of FIG. 1;
- (4) FIG. 4 is a partial perspective exploded view of FIG. 3 from another angle;
- (5) FIG. 5 is a perspective view after removing a housing and magnets in FIG. 3;
- (6) FIG. 6 is a perspective exploded view of FIG. 5;
- (7) FIG. 7 is a perspective exploded view of FIG. 6 from another angle;
- (8) FIG. 8 is a top view of partial elements of the electrical connector assembly of FIG. 3 when the partial elements are assembled together, wherein a connector rotating unit is located in an original/normal position; and
- (9) FIG. 9 is atop view of FIG. 8 from another angle, wherein the connector rotating unit is located in a rotated position.

DETAILED DESCRIPTION

(10) Exemplary embodiments will be described in detail here, examples of which are shown in drawings. When referring to the drawings below, unless otherwise indicated, same numerals in different drawings represent the same or similar elements. The examples described in the following exemplary embodiments do not represent all embodiments consistent with this application. Rather, they are merely examples of devices and methods consistent with some aspects of the application as detailed in the appended claims.

(11) The terminology used in this application is only for the purpose of describing particular embodiments, and is not intended to limit this application. The singular forms “a”, “said”, and “the” used in this application and the appended claims are also intended to include plural forms unless the context clearly indicates other meanings.

(12) It should be understood that the terms “first”, “second” and similar words used in the specification and claims of this application do not represent any order, quantity or importance, but are only used to distinguish different components. Similarly, “an” or “a” and other similar words do not mean a quantity limit, but mean that there is at least one; “multiple” or “a plurality of” means two or more than two. Unless otherwise noted, “front”, “rear”, “lower” and/or “upper” and similar words are for ease of description only and are not limited to one location or one spatial orientation. Similar words such as “include” or “comprise” mean that elements or objects appear before “include” or “comprise” cover elements or objects listed after “include” or “comprise” and their equivalents, and do not exclude other elements or objects. The term “a plurality of” mentioned in the present disclosure includes two or more.

(13) Hereinafter, some embodiments of the present disclosure will be described in detail with reference to the accompanying drawings. In the case of no conflict, the following embodiments and features in the embodiments can be combined with each other.

(14) Referring to FIG. 1 to FIG. 9, an illustrated embodiment of the present disclosure discloses an electrical connector assembly, which includes a housing 1, a connector rotating unit 2, a plurality of magnets and a cable 4. In the illustrated embodiment of the present disclosure, the housing 1 includes an outer housing 10, a first housing 11, a second housing 12, and a side housing 13. The first housing 11 is mated with the second housing 12. Both the first housing 11 and the second housing 12 are located in the outer housing 10. The side housing 13 is at least partially inserted into

the outer housing **10** from one side and locked with the outer housing **10**.

(15) The housing **1** defines a slot **100** extending in a first direction **A1-A1**. In the illustrated embodiment of the present disclosure, part of the slot **100** is provided in the outer housing **10**, and another part of the slot **100** is provided in the side housing **13**. In other words, the slot **100** is jointly formed by the outer housing **10** and the side housing **13**.

(16) Referring to FIG. 3 and FIG. 4, the first housing **11** includes a first accommodating space **111** and two first arc-shaped supporting portions **112** located on two sides of the first accommodating space **111**, respectively. The two first arc-shaped supporting portions **112** are exposed in the first accommodating space **111**.

(17) Similarly, the second housing **12** includes a second accommodating space **121** and two second arc-shaped supporting portions **122** located on two sides of the second accommodating space **121**, respectively. The two second arc-shaped supporting portions **122** are exposed in the second accommodating space **121**.

(18) The connector rotating unit **2** is rotatable relative to the housing **1** under the action of an external force. In the embodiment shown in the present disclosure, the connector rotating unit **2** includes a rotating body **21**, a movable post **25** and an electrical connector **23** fixedly connected to the rotating body **21**. The electrical connector **23** passes through the slot **100** in a second direction **A2-A2** perpendicular to the first direction **A-A1** so as to at least partially extend beyond the housing **1**. Referring to FIG. 1 and FIG. 2, the slot **100** includes a first groove **101** on one side of the electrical connector **23** and a second groove **102** on another side of the electrical connector **23**. The first groove **101** and the second groove **102** are configured to provide a space for the electrical connector **23** to rotate left and right.

(19) It is understandable to those skilled in the art that, in an embodiment of the present disclosure, a pivot (not shown) is provided between the rotating body **21** and the housing **1**. The rotating body **21** is rotatable relative to the housing **1** with the pivot as a center of rotation. Of course, it is understandable to those of ordinary skill in the art that, in other embodiments, the rotating body **21** can also rotate relative to the housing **1** by means of other structural designs, which will not be repeated in the present disclosure.

(20) The electrical connector assembly is further provided with an arc-shaped limiting groove **240** mated with the movable post **25** and an elastic member mated with the movable post **25**. One end of the elastic member is fixed relative to the housing **1**, and the other end of the elastic member is connected to the movable post **25**. The movable post **25** is movable in the arc-shaped limiting groove **240**. The elastic member is configured to drive the movable post **25** by a restoring force of the elastic member after the connector rotating unit **2** is rotated, so as to rotate the connector rotating unit **2** back to its original/normal position.

(21) Referring to FIG. 5 to FIG. 7, the rotating body **21** is generally of a D-shaped configuration, which includes an arc surface **211**, a flat surface **212**, a receiving slot **213** extending through the arc surface **211**, and an insertion hole **214** extending through the rotating body **21**. In the embodiment shown in the present disclosure, the movable post **25** is fixed in the insertion hole **214**, so that the movable post **25** can rotate together with the rotating body **21**. The rotating body **21** includes a first arc-shaped groove **215** mated with the first arc-shaped supporting portions **112**, a first abutting surface **216** exposed in the first arc-shaped groove **215**, a second arc-shaped groove **217** mated with the second arc-shaped supporting portions **122**, and a second abutting surface **218** exposed in the second arc-shaped groove **217**. After assembly, the rotating body **21** is located between the first housing **11** and the second housing **12**. The rotating body **21** is accommodated in the first accommodating space **111** and the second accommodating space **121**. In the embodiment shown in the present disclosure, the first arc-shaped supporting portion **112** mates with the first abutting surface **216** to prevent the rotating body **21** from moving upwardly. The second arc-shaped supporting portion **122** mates with the second abutting surface **218** to prevent the rotating body **21** from moving downwardly. However, the rotating body **21** can still rotate in a horizontal plane.

(22) In the illustrated embodiment of the present disclosure, the electrical connector assembly includes a built-in circuit board **24** and a cable **4**. The arc-shaped limiting groove **240** is formed on the built-in circuit board **24**. The cable **4** includes a plurality of conductive portions **41**. The plurality of conductive portions **41** are electrically connected to the built-in circuit board **24**. Specifically, in the embodiments shown in the present disclosure, the built-in circuit board **24** includes a first surface **241**, a second surface **242** opposite to the first surface **241**, and a counterbore area **243** recessed from the first surface **241** toward the second surface **242**. The counterbore area **243** is provided with a mounting hole **2431** extending through the second surface **242**. The arc-shaped limiting groove **240** is located adjacent to the counterbore area **243**. The built-in circuit board **24** further includes a plurality of arc-shaped conductive pads **244** arranged at intervals. In the embodiment shown in the present disclosure, the plurality of arc-shaped conductive pads **244** are located on the first surface **241** and the second surface **242** of the built-in circuit board **24**. For the plurality of arc-shaped conductive pads **244** located on the same side, arc lengths of the plurality of arc-shaped conductive pads **244** decrease sequentially along a direction away from the arc-shaped limiting groove **240**. In the embodiment shown in the present disclosure, the counterbore area **243** includes a first limiting side surface **2432** on one side and a second limiting side surface **2433** on an opposite side. The first limiting side surface **2432** and the second limiting side surface **2433** form an included angle. The mounting hole **2431** is approximately located at an apex of the included angle. The arc-shaped limiting groove **240** is curved toward a direction of the mounting hole **2431**.

(23) In the illustrated embodiment of the present disclosure, the electrical connector **23** is a connector plug (e.g., a Universal Serial Bus (USB) Type C plug), which includes a plurality of conductive terminals **231** corresponding to the plurality of arc-shaped conductive pads **244**. Each of the plurality of conductive terminals **231** is always in contact with a corresponding arc-shaped conductive pad **244** no matter whether the connector rotating unit **2** rotates or not. The plurality of conductive terminals **231** protrude toward the built-in circuit board **24** in a cantilever shape.

Specifically, the plurality of conductive terminals **231** include a plurality of cantilever portions **2313** protruding toward the built-in circuit board **24**. Each cantilever portion **2313** is provided with a dimple **2313a** (see FIG. 7) which is in contact with the corresponding arc-shaped conductive pad **244**. In the embodiment shown in the present disclosure, the plurality of conductive terminals **231** include a first row of conductive terminals **2311** and a second row of conductive terminals **2312**.

(24) The built-in circuit board **24** is clamped between the first row of conductive terminals **2311** and the second row of conductive terminals **2312**. Regardless of the position of the connector rotating unit **2**, the first row of conductive terminals **2311** and the second row of conductive terminals **2312** are always in contact with the built-in circuit board **24**. The built-in circuit board **24** is received in the receiving slot **213** of the rotating body **2**. The built-in circuit board **24** is fixed relative to the housing **1**, that is, the built-in circuit board **24** will not rotate with the rotation of the connector rotating unit **2**.

(25) In the illustrated embodiment of the present disclosure, the elastic member is a tension spring **6**. The electrical connector assembly is provided with a fixing post **26** which is fixed in the mounting hole **2431** of the built-in circuit board **24**. One end of the tension spring **6** is fixed to the fixing post **26**, and the other end of the tension spring **6** is connected to the movable post **25**.

(26) Specifically, referring to FIG. 5 to FIG. 9, in the embodiment illustrated in the present disclosure, the tension spring **6** includes a first annular portion **61** located at one end of the tension spring **6** and a second annular portion **62** located at the other end of the tension spring **6**. The first annular portion **61** is sleeved on the fixing post **26**. The second annular portion **62** is sleeved on the movable post **25**. The tension spring **6** is located at least partially in the counterbore area **243**. The tension spring **6** is limited between the first limiting side surface **2432** and the second limiting side surface **2433** when the connector rotating unit **2** rotates.

(27) The arc-shaped limiting groove **240** includes a stop recess **2401** at the top thereof. After the

connector rotating unit **2** is rotated from a rotated position back to the original/normal position, the movable post **25** is at least partially positioned in the stop recess **2401**. It is understandable to those skilled in the art that, in the embodiment shown in the present disclosure, the stop recess **2401** deviates outward from other portions of the arc-shaped limit groove **240**. That is, the stop recess **2401** and other portions of the arc-shaped limiting groove **240** are not smoothly communicated. The design advantage of the stop recess **2401** is that it has a certain foolproof effect when the connector rotating unit **2** is rotated back to the original/normal position. That is, the movable post **25** needs a relatively large force to realize a second rotation.

(28) The plurality of magnets include a first magnet **33** and a second magnet **34** disposed in the housing **1**. The first magnet **33** and the second magnet **34** are located on two sides (e.g., left and right sides) of the rotating body **21**, respectively. The first magnet **33** and the second magnet **34** are configured to be attracted to a magnetic attraction element (e.g., a magnet or an iron) of a mating device, so as to improve the mating reliability. It should be noted that the technical term “magnetic attraction element” used in the present disclosure is intended to indicate that the element can attract or repel a corresponding element by utilizing the principle of a magnetic field.

(29) The electrical connector assembly further includes another electrical connector **7** connected to the cable **4**. The another electrical connector **7** is electrically connected to the electrical connector **23** via the cable **4** and the built-in circuit board **24**. In the embodiment of the present disclosure, the another electrical connector **7** may be a USB Type C plug, a USB Type A plug, a Micro USB plug, a High Definition Multimedia Interface (HDMI) plug, a DisplayPort plug, a Thunderbolt plug, or the like.

(30) Referring to FIG. **8** and FIG. **9**, during use, when an external force acting on the connector rotating unit **2** overcomes an engaging force between the movable post **25** and the stop recess **2401**, the connector rotating unit **2** is driven to rotate by the external force. The movable post **25** moves in the arc-shaped limiting groove **240**. The tension spring **6** is stretched to generate a certain restoring force. When the external force disappears, the restoring force of the tension spring **6** drives the movable post **25** automatically to return to the stop recess **2401**, thereby rotating the connector rotating unit **2** back to the original/normal position. As a result, the user experience is greatly improved.

(31) The above embodiments are only used to illustrate the present disclosure and not to limit the technical solutions described in the present disclosure. The understanding of this specification should be based on those skilled in the art. Descriptions of directions, although they have been described in detail in the above-mentioned embodiments of the present disclosure, those skilled in the art should understand that modifications or equivalent substitutions can still be made to the application, and all technical solutions and improvements that do not depart from the spirit and scope of the application should be covered by the claims of the application.

Claims

1. An electrical connector assembly, comprising: a housing; a connector rotating unit, the connector rotating unit being rotatable under an action of an external force, the connector rotating unit comprising a rotating body, a movable post and an electrical connector fixedly connected with the rotating body, the electrical connector at least partially extending beyond the housing; a board, the board being secured to the housing, and the board defining an arc-shaped limiting groove mated with the movable post; a fixing post, the fixing post being secured to the board; and an elastic member mated with the fixing post and the movable post; wherein one end of the elastic member is mounted to the fixing post, and another end of the elastic member is connected to the movable post; the movable post is movable in the arc-shaped limiting groove; the elastic member is configured to drive the movable post by a restoring force of the elastic member after the connector rotating unit is rotated, so as to rotate the connector rotating unit back to an original position.

2. The electrical connector assembly according to claim 1, wherein the elastic member is a tension spring, one end of the tension spring is fixed to the fixing post, and another end of the tension spring is connected to the movable post.
3. The electrical connector assembly according to claim 2, wherein the tension spring comprises a first annular portion located at one end of the tension spring and a second annular portion located at another end of the tension spring; and wherein the first annular portion is sleeved on the fixing post, and the second annular portion is sleeved on the movable post.
4. The electrical connector assembly according to claim 2, wherein the board is a built-in circuit board.
5. The electrical connector assembly according to claim 4, wherein the built-in circuit board comprises a first surface, a second surface opposite to the first surface, and a counterbore area recessed from the first surface toward the second surface, the counterbore area is provided with a mounting hole extending through the second surface, the mounting hole is configured for mounting the fixing post, and the arc-shaped limiting groove is located adjacent to the counterbore area.
6. The electrical connector assembly according to claim 4, wherein the built-in circuit board further comprises a plurality of arc-shaped conductive pads arranged at intervals, the electrical connector comprises a plurality of conductive terminals corresponding to the plurality of arc-shaped conductive pads, the plurality of conductive terminals are always in contact with the plurality of arc-shaped conductive pads when the connector rotating unit rotates.
7. The electrical connector assembly according to claim 6, wherein arc lengths of the plurality of arc-shaped conductive pads decrease sequentially along a direction away from the arc-shaped limiting groove, each conductive terminal comprises a cantilever portion protruding toward the built-in circuit board, and each cantilever portion is provided with a dimple in contact with a corresponding arc-shaped conductive pad.
8. The electrical connector assembly according to claim 1, further comprising a first magnet and a second magnet; wherein the first magnet and the second magnet are mounted to the housing, and are configured to be attracted to a magnetic attraction element of a mating device.
9. The electrical connector assembly according to claim 4, further comprising a cable having a plurality of conductive portions; wherein the plurality of conductive portions are electrically connected to the built-in circuit board; the cable extends out of the housing.
10. The electrical connector assembly according to claim 1, wherein the arc-shaped limiting groove comprises a stop recess at a top thereof; and wherein after the connector rotating unit is rotated back to the original position, the movable post is at least partially positioned in the stop recess.
11. An electrical connector assembly, comprising: a housing comprising a first housing and a second housing, the first housing defining a first accommodating space, the second housing defining a second accommodating space; and a connector rotating unit, the connector rotating unit being rotatable under an action of an external force, the connector rotating unit comprising a rotating body, a movable post and an electrical connector fixedly connected with the rotating body, the rotating body being at least partially received in the first accommodating space and the second accommodating space, the electrical connector at least partially extending beyond the housing; a board, the board being secured to the housing, and the board defining a limiting groove; a fixing post, the fixing post being secured to the board; and an elastic member at least partially disposed between the first housing and the second housing; wherein one end of the elastic member is mounted to the fixing post, and another end of the elastic member is connected to the movable post; the movable post is movable in the limiting groove when the connector rotating unit is rotating; the elastic member is configured to drive the movable post by a restoring force of the elastic member so as to rotate the connector rotating unit back to an original position.
12. The electrical connector assembly according to claim 11, wherein the elastic member is a tension spring, one end of the tension spring is fixed to the fixing post, and another end of the tension spring is connected to the movable post.

13. The electrical connector assembly according to claim 12, wherein the tension spring comprises a first annular portion located at one end of the tension spring and a second annular portion located at another end of the tension spring; and wherein the first annular portion is sleeved on the fixing post, and the second annular portion is sleeved on the movable post.
14. The electrical connector assembly according to claim 12, wherein the board is a built-in circuit board.
15. The electrical connector assembly according to claim 14, wherein the built-in circuit board comprises a first surface, a second surface opposite to the first surface, and a counterbore area recessed from the first surface toward the second surface, the counterbore area is provided with a mounting hole extending through the second surface, the mounting hole is configured for mounting the fixing post, and the limiting groove is located adjacent to the counterbore area.
16. The electrical connector assembly according to claim 14, wherein the built-in circuit board further comprises a plurality of arc-shaped conductive pads arranged at intervals, the electrical connector comprises a plurality of conductive terminals corresponding to the plurality of arc-shaped conductive pads, the plurality of conductive terminals are always in contact with the plurality of arc-shaped conductive pads when the connector rotating unit rotates.
17. The electrical connector assembly according to claim 16, wherein arc lengths of the plurality of arc-shaped conductive pads decrease sequentially along a direction away from the limiting groove, each conductive terminal comprises a cantilever portion protruding toward the built-in circuit board, and each cantilever portion is provided with a dimple in contact with a corresponding arc-shaped conductive pad.
18. The electrical connector assembly according to claim 14, further comprising a cable having a plurality of conductive portions; wherein the plurality of conductive portions are electrically connected to the built-in circuit board; the cable extends out of the housing.
19. The electrical connector assembly according to claim 11, wherein the limiting groove is of an arc-shaped configuration and comprises a stop recess at a top thereof; and wherein after the connector rotating unit is rotated back to the original position, the movable post is at least partially positioned in the stop recess.
20. The electrical connector assembly according to claim 11, wherein the first housing comprises two first arc-shaped supporting portions located on two sides of the first accommodating space, respectively; the second housing comprises two second arc-shaped supporting portions located on two sides of the second accommodating space, respectively; the rotating body comprises a first arc-shaped groove mating with the first arc-shaped supporting portions and a second arc-shaped groove mating with two second arc-shaped supporting portions.
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